USER MANUAL

Accessory 34AA

Opto 32-Bit Input/ 32-Bit Output Board

3Ax-602817-xUxx

July 6, 2004

DELTA TAU
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INTRODUCTION

The Accessory 34AA(ACC-34AA) for PMAC and Turbo PMAC controllers is a general-purpose discrete input/output (I/O) board. ACC-34AA provides 32 lines of optically isolated inputs and 32 lines of optically isolated outputs. Port A contains the 32 input lines (AI0 to AI31). Port B contains the 32 output lines (BO0 to BO31). The actual I/O Reads and Writes are carried out using a special form of M-variables, which will be described later in this manual. ACC-34AA is one of a series of I/O accessories for PMAC that uses the JTHW multiplexer port connector. Others are:

ACC-34A	The Opto 64-bit Input/Output board
ACC-34B	The 64-bit Standard Opto I/O Interface board
ACC-18	The Thumbwheel Multiplexer board
ACC-8D Option 7	The Resolver to Digital Converter board
ACC-8D Option 8	The Yaskawa® Absolute Encoder Interface board

All of the above accessories use the JTHW multiplex address scheme, and several of them may be daisy-chained to a single PMAC. For enhanced noise reduction and long distance installation, Accessories 35A and 35B provide differential buffer capabilities for the JTHW signals. The use of the long distance buffer pair (ACC-35A and ACC-35B) is recommended whenever the required cable length between PMAC and ACC-34x is over 3 meters (10 feet).

Up to 32 ACC-34AAs may be connected to a single PMAC, which gives a possible 1,024 input and 1,024 output lines in addition to those available on the PMAC board and on the parallel I/O expansion boards. Accessory 34 communicates to PMAC via its JTHW connector.

The ACC-34AA is compatible operationally with the older ACC-34A board, and provides several important improvements.

The improvements that the ACC-34AA has with respect to the ACC-34A are:

- Individual optical isolation of all I/O points
- RC filter with 1 msec time constant on all inputs
- Parity checking on serial communications with PMAC
- Two multiplexer port headers for easier daisychaining

Parity Checking

Parity checking is done on all serial communications between the PMAC and the ACC-34AA. This requires PMAC firmware version 1.16 or newer.

When PMAC sends data to the ACC-34AA 32-bit output word by writing to the TWS M-variable pointing to that word, the ACC-34AA evaluates the parity bits sent with that word and compares them to the parity bits it calculated itself. If there is a difference, the output word is ignored (the outputs stay in the state of the last successful write), and PMAC is notified of a parity error. PMAC shows this error by setting bit 6 of the global status register X:$0003 (X:$000006 in Turbo PMAC). The user program can check this bit with an M-variable to see if the data must be sent again.

When the ACC-34AA sends the 32-input word to PMAC by a PMAC read of the TWS M-variable pointing to that word, the ACC-34AA creates parity bits that are sent with the word. PMAC evaluates the parity bits sent with that word and compares them to the parity bits it calculated itself. If there is a difference, PMAC shows the error by setting this parity error bit. The user program can check this bit with an M-variable to see if the data must be sent again.

Optical Isolation

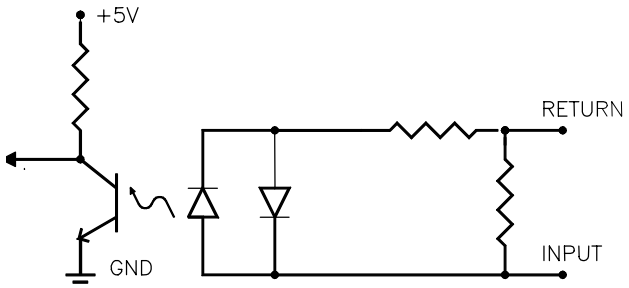
If the optical isolation on the ACC-34AA is not to be defeated, use different power supplies for the ACC-34AA's internal circuitry, via terminal block TB9, and for the I/O points themselves, via TB1 through TB8. Each I/O terminal block can have a separate isolated supply.

Port Headers

The ACC-34AA has two 26-pin headers, J1A and J1B, that can be used for connection of boards to the PMAC JTHW Multiplexer port (ACC-34A has a single J1 header). The identical signals are present on both connectors with a simple pass-through on the board, so it is possible to use the second header to connect to the next accessory board daisychained to the multiplexer port, instead of a multi-drop cable.

ACC-34AA HARDWARE SPECIFICATIONS

The inputs to the ACC-34AA board have an activation range from 12V to 24V, and can be either sinking or sourcing depending on the reference to the opto circuitry. The opto-isolator IC used is a PS2705 NEC phototransistor output type. This IC allows the current to flow from return to flag (sinking) or from flag to return (sourcing).

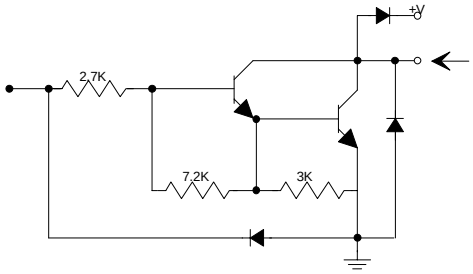


ACC-34AA output drivers are organized in a set of four 8-bit groups. Each group (each byte) may be ordered with either current sourcing drivers (default) or with current sinking drivers. The default configuration of this accessory board uses UDN2981 current sourcing drivers for the four 8-bit output groups. With this configuration, the current drawn from each output line is limited to 100mA at voltage levels between 12 and 24V.

Custom orders (ordered as ACC-34AA Option 1) are available for current sinking applications. In current sinking configurations, one ULN2803 driver is used for each 8-bit output group. Each open collector output line can sink up to 100mA when pulled up to a voltage level between 12 to 24V (external pull-up resistors are not supplied). When this accessory is ordered with its Option 1, the mixing of current sourcing and sinking drivers is possible (in 8-bit groups).

Sinking Outputs:

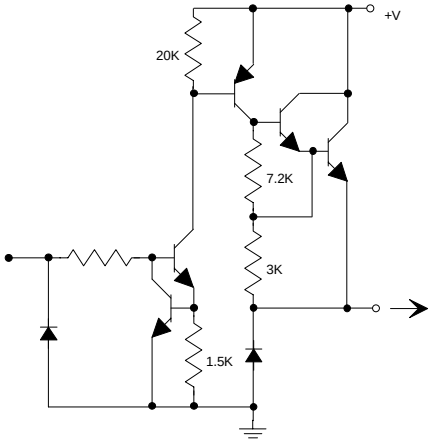
OUTPUT CHIP EQUIVALENT
CIRCUIT ULN2803 FOR SINKING



INVERTING, OPEN COLLECTOR, SINKING, 12-24V

Sourcing Outputs:

OUTPUT CHIP EQUIVALENT
CIRCUIT UDN2981 FOR SOURCING



NON-INVERTING, SOURCING, 12-24V

Power Supply and Opto-Isolation Considerations

The power for the PMAC processor side of the opto-isolation circuitry is brought in directly from J1A or J1B (JTHW). The power for the external side of the opto-isolation circuitry should be from a separate supply brought in through TB9. This can be any level from 12V to 24V. The current requirements for the power brought in through TB9 is approximately 0.4A at 24V or 0.8A at 12V.

For TB1 to TB4 inputs, the power supply Ω can range from 12 to 24V. At 24V, the current requirement is approximately 70mA.

For TB5 to TB8 outputs in the sourcing configuration, the supply voltage may range from 12 to 24V. The maximum current requirement should be limited to 100mA per output line, which translates to 800mA per terminal block. For sinking configurations, the supply voltage may range between 12 and 24V (the maximum current is limited to that which flows through the nine 3.3 K Ω pull-up resistors (approximately 70mA).

ACC-34AA LED Indicators

The ACC-34AA is equipped with bi-color LEDs allowing verification of a sinking or sourcing signal. On power up to the board, the low bit LED will display green if the jumpers for the I/O port are set for sinking. It will display red if set for sourcing. (See ACC-34AA Hardware Reference manual for exact jumper settings.) Similarly, the actual I/O LEDs will display green when pulled low or red when pulled high.

ACC-34AA Watchdog Timer

Watchdog timer circuitry is enabled on ACC-34AA when jumper E22 is not installed. This jumper is installed in the default factory setup. When the timer is enabled and if PMAC (through a PLC program) does not read from or write to the ACC-34AA board at least once per 1.5 seconds, the watchdog timer trips. When this occurs, the output transistors are turned OFF (no current is driven on any of the 32 outputs). A subsequent read or write always reinitializes the board.

J1A and J1B

This 26-pin Header provides the link between PMAC's JTHW (multiplexer port connector) and ACC-34AA. Using the supplied flat cable, connect PMAC to J1A or J1B. Through this connector, PMAC sets the outputs and reads the inputs. The power for the processor side of the opto-isolation circuitry is provided from the PMAC board through this connector. For the connection of multiple ACC-34AAs to a single PMAC, and additional cable is required for each ACC-34AA.

TB1

This 10-pin terminal block is used to connect to the first set of eight input lines (AI0 to AI7). This connector (12 to 24V, 70mA) is used to bring in a separate power supply for the pull-up/down resistors of AI0 to AI7.

TB2

This 10-pin terminal block is used to connect to the second set of eight input lines (AI8 to AI15). In addition, this connector (12 to 24V, 70mA) is used to bring in a separate power supply for the pull-up/down resistors of AI8 to AI15.

TB3

This 10-pin terminal block is used to connect to the third set of eight input lines (AI16 to AI23). This connector (12 to 24V, 70mA) is used to bring in a separate power supply for the pull-up/down resistors of AI16 to AI23.

TB4
This 10-pin terminal bloc1k is used to connect to the fourth set of eight input lines (AI24 to AI31). This connector (12 to 24V, 70mA) is used to bring in a separate power supply for the pull-up/down resistors of AI24 to AI31.

TB5
This 10-pin terminal block is used to connect to the first set of eight output lines (BO0 to BO7). This connector is used to bring in a separate power supply for BO0 to BO7. In the sourcing configuration, this supply will be used to drive the loads on the eight lines (up to 800mA maximum at 24V). In the sinking configuration, the current requirement from this power supply is minimal (approximately 70mA).

TB6
This 10-pin terminal block is used to connect to the second set of eight output lines (BO8 to BO15). This connector is used to bring in a separate power supply for BO8 to BO15. In the sourcing configuration, this supply will be used to drive the loads on all eight lines (up to 800mA maximum at 24V). In the sinking configuration, the current requirement from this power supply is minimal (approximately 70mA).

TB7
This 10-pin terminal block is used for the connection to the third set of eight output lines (BO16 to BO23). This connector is used to bring in a separate power supply for BO16 to BO23. In the sourcing configuration, this supply will be used to drive the loads on all the eight lines (up to 800mA maximum at 24V). In the sinking configuration, the current requirement from this power supply is minimal (approximately 70mA).

TB8
This 10-pin terminal block is used to connect to the fourth set of eight output lines (BO24 to BO31). This connector is used to bring in a separate power supply for BO24 to BO31. In the sourcing configuration, this supply will be used to drive the loads on all eight lines (up to 800mA maximum at 24V). In the sinking configuration, the current requirement from this power supply is minimal (approximately 70mA).

TB9
This is a 2-pin terminal block used to bring in the power supply (12-24V) for logic circuits on the I/O side of the opto-isolation circuitry. For proper operation of the board, this power supply must be brought in through TB9.

ACC-34AA Options
There are three options for the ACC-34AA: Either Option 1 or Option 2 must be selected.

Option1
Sinking Outputs

Option2
Sourcing Outputs

Option3
Din Rail Mount

Jumper Specifications

E1-E8*: Sinking or Sourcing Input Select

Jumpers	Descriptions
E1 & E2	1-2 Sinking inputs at TB1 2-3 Sourcing inputs at TB1
E3 & E4	1-2 Sinking inputs at TB2 2-3 Sourcing inputs at TB2
E5 & E6	1-2 Sinking inputs at TB3 2-3 Sourcing inputs at TB3
E7 & E8	1-2 Sinking inputs at TB4 2-3 Sourcing inputs at TB4
* Set by factory	

E9-E16*: Sinking or Sourcing Output Select

Jumpers	Descriptions
E9 & E10	1-2 Sinking outputs with the ULN2803A IC for TB5 outputs 2-3 Sourcing outputs with the UDN2981A IC for TB5 outputs
E11 & E12	1-2 Sinking outputs with the ULN2803A IC for TB6 outputs 2-3 Sourcing outputs with the UDN2981A IC for outputs
E13 & E14	1-2 Sinking outputs with the ULN2803A IC for TB7 outputs 2-3 Sourcing outputs with the UDN2981A IC TB7 for outputs
E15 & E16	1-2 Sinking outputs with the ULN2803A IC for TB8 outputs 2-3 Sourcing outputs with the UDN2981A IC for TB8 outputs
* Set by factory	

E22: ACC-34AA Watchdog Timer Jumper

Jumpers	Descriptions	Default
E22	On - ACC-34AA Watchdog timer is disabled Off – ACC-34AA Watchdog timer is enabled	On

E23: ACC-34AA Standard/Alternate Read

Jumpers	Descriptions	Default
E23	1-2 - Normal Multiplex port read 2-3 - Custom operation (not supported by Delta Tau Data Systems, Inc.)	1-2

Multiplex Address Map

Each ACC-34 occupies eight bytes of address space on the PMAC's JTHW multiplex memory space. This memory space is eight bits wide, which provides the ability to daisychain 32 (256/8) ACC-34xs together (or a combination of ACC-34xs, ACC-18s and ACC-8D OPT7s). The 5-bit DIP switch SW1, determines the address of each ACC-34x board on the allocated memory space. Port A occupies the base address (i.e., bytes 0, 8, 16, etc.) and Port B occupies the base address plus four (i.e., bytes 4, 12, 20, etc.).

SW1 DIP Switch Setting

The following table shows how SW1 should be set for one or more ACC-34x boards connected to the same PMAC.

Board #	Byte (Port A & Port B)	5	4	3	2	1
#1	0 & 4	ON	ON	ON	ON	ON
#2	8 & 12	ON	ON	ON	ON	OFF
#3	16 & 20	ON	ON	ON	OFF	ON
#4	24 & 28	ON	ON	ON	OFF	OFF
#5	32 & 36	ON	ON	OFF	ON	ON
#6	40 & 44	ON	ON	OFF	ON	OFF
#7	48 & 52	ON	ON	OFF	OFF	ON
#8	56 & 60	ON	ON	OFF	OFF	OFF
#9	64 &68	ON	OFF	ON	ON	ON
#10	72 & 76	ON	OFF	ON	ON	OFF
#11	80 & 84	ON	OFF	ON	OFF	ON
#12	88 &92	ON	OFF	ON	OFF	OFF
#13	96 & 100	ON	OFF	OFF	ON	ON
#14	104 & 108	ON	OFF	OFF	ON	OFF
#15	112 & 116	ON	OFF	OFF	OFF	ON
#16	120 & 124	ON	OFF	OFF	OFF	OFF
#17	128 & 132	OFF	ON	ON	ON	ON
#18	136 & 140	OFF	ON	ON	ON	OFF
#19	144 & 148	OFF	ON	ON	OFF	ON
#20	152 & 156	OFF	ON	ON	OFF	OFF
#21	160 & 164	OFF	ON	OFF	ON	ON
#22	168 & 172	OFF	ON	OFF	ON	OFF
#23	176 & 180	OFF	ON	OFF	OFF	ON
#24	184 &188	OFF	ON	OFF	OFF	OFF
#25	192 & 196	OFF	OFF	ON	ON	ON
#26	200 & 204	OFF	OFF	ON	ON	OFF
#27	208 & 212	OFF	OFF	ON	OFF	ON
#28	216 & 220	OFF	OFF	ON	OFF	OFF
#29	124 & 228	OFF	OFF	OFF	ON	ON
#30	232 & 236	OFF	OFF	OFF	ON	OFF
#31	240 & 244	OFF	OFF	OFF	OFF	ON
#32	248 & 252	OFF	OFF	OFF	OFF	OFF

This table shows the daisy-chain board address relationship with respect to the 5-bit (SW1) DIP position setting.
Note: ON=Closed, OFF=Open. To turn off a switch, push down on the open side. To turn on a switch, push down on the numbered side.

INPUT AND OUTPUT PORTS

Port A is always configured as an input port, and the output lines are driven by writing to Port B. The input and the output lines can be read or written using TWS type M-variables. Once the correct M-variable definition assignment is done, use these M-variables in the usual ways (e.g., motion programs, PLC programs, etc.).

Jumpers E1 through E8 determine the sinking/sourcing characteristics of the input port. For sourcing a high level on, an input line is read as binary 1 (positive logic). For sinking a high level on, an input line is read as binary 0 (negative logic).

Jumpers E9 through E16 together with drivers U51 through U54 determine the sinking/sourcing characteristic of the output port. For a sourcing setup, writing a binary 1 generates a high level on the corresponding output line (positive logic). For a sinking setup, writing a binary 1 turns on the corresponding open collector transistor output, and this in turn switches on the output (negative logic).

M-Variable Assignments

TWS is a special format 32-bit wide M-variable for reading the data from, and writing the data to an ACC-34 card:.

```
M{constant}->TWS:{m-plex}          ;Serial Thumbwheel Multiplexer M-
                                   ;Variable Definition
```

This command causes PMAC to define the specified M-variable to point to a 32-bit word of input or output serially multiplexed on the thumbwheel port on an ACC-34x board.

Note

The individual bits of the thumbwheel port on an ACC-34x board cannot be directly assigned to an M-variable. Only 32-bit words (ports) of input or output can be accessed.

The address on the multiplex port specified here must match the address set by the DIP switch on the ACC-34x board. The ACC-34x Hardware Reference manual contains a table listing all of the possibilities.

The best procedure for using TWS M-variables in a program is as follows: The input word (TWS M-variable) should be copied into its image variable at the beginning of a sequence of operations. The operations can then be done on the image variable without requiring PMAC to actually read or write to the I/O port for each operation. The output word is first assembled into its image variable, and then copied to the actual output word once at the end of a sequence of operations.

This procedure will allow the most efficient and flexible use of TWS M-variables.

Note

This type of variable can only be used in background tasks (PLCs and PLCCs 1-31). They cannot be used in foreground tasks (motion programs, PLC and PLCC 0).

For an input port, {m-plex} is a legal byte number (from column 2 of Table 1) plus one. Any attempt to write to a TWS type M-variable defined with bit zero of its address set to one, is automatically prevented by PMAC's firmware. For an output port, {m-plex} is a legal byte number (from column 2 of Table 1) plus two. An attempt to read a TWS type M-variable defined with bit one of its address set to one, returns zero, and the actual read is prevented by PMAC's firmware.

Example: To address Port A (bits 0 to 31) of board #1 as an input using M100, use the following definition:

Board #	Byte(Port A & Port B)	5	4	3	2	1

#1	0 & 4	ON	ON	ON	ON	ON

```
M100->TWS:1  ; Port A (AIO 0-31) of an ACC-34 with SW1 switches all ON
             ; assigned for read only (1=0+1)
```

Similarly, to address Port B of the same board #1 as an output using M101, use the following definition:

```
M101->TWS:6  ; Port B (BIO 0-31) of an ACC-34 with SW1 switches all ON
             ; assigned for write only (4+2=6)
```

To address Port A of board #6 as an input using M300, use the following definition:

Board #	Byte(Port A & Port B)	5	4	3	2	1
#6	40 & 44	ON	ON	OFF	ON	OFF

```
M300->TWS:41  ; Port A (AIO 0-31) of an ACC-34 with SW1 switches
              ; assigned for read only (41=40+1)
```

Note

A 32-bit Read or a 32-bit Write to an individual port takes approximately 64 microseconds of time in the PMAC's background time slot (20MHz CPU, significantly less on faster CPUs). As a result, excessive and unnecessary references to TWS-type M-variables is not recommended.

Processing ACC-34x Inputs & Outputs

Because the PMAC interface to the Accessory 34 family of I/O boards (ACC-34x) is by full 32-bit words transmitted serially (even when access to only a single bit is desired), carefully consider how the interface is done and how frequently. Care must also be taken to work efficiently with the data so that PMAC is not bogged down with slow serial reads and writes, and time-consuming logic to assemble and disassemble I/O words.

The recommended strategy is to keep images of each input or output word in PMAC's internal memory, or in the dual-ported RAM. The input words are copied into their image words, and the output words are copied from their image words. Most program operations deal with these image words; this way, slow transfer to or from an ACC-34x board is performed less frequently. During the act of copying, bit inversion can also be performed with the exclusive-or function.

When to Access ACC-34x

The actual reads and writes for an ACC-34x board can be done only in a background PLC program (PLC 1-31) or through on-line commands, which are executed between PLC programs. Motion programs and PLC 0 cannot access this I/O directly. They can work only with the image words. Reading an input word from an ACC-34x is simply a question of using the TWS-form M-variable for that word on the right side of an equation. Usually, this operation simply copies the input word into its internal image variable. Similarly, writing an output word to an ACC-34x involves using the M-variable for that word on the left side of an equation, typically just copying it from its internal image word.

Most will treat ACC-34x I/O the same way that a traditional PLC treats its I/O; all of the inputs are read at the beginning of a PLC software scan, and all of the outputs are written to at the end of the scan. In between, all the processing of the variables is done while working with the internal image words. It is possible to make the write operation to the output word conditional on a change in the image word for the output from the previous scan, but the time involved in making the decision and storing each scan's value is about the same as the actual writing to the output.

ACC-34X SETUP

For the purpose of ACC-34x setup, the following example will demonstrate how to utilize 32 inputs and 32 outputs of an ACC-34x. The following three variables will be used during the ACC-34x I/O procedure:

Actual Word Variable	Variable which is read or written to by ACC-34x & PMAC
Image Word Variable	Variable assigned set equal to (Image) actual word variable
Image Bit Variable	Single bit of image word variable

Image Word Variables

It is best to use fixed-point M-variables as the internal image variables for the I/O words. When this is done, a single M-variable representing the entire I/O word can be used for the copying operation. Then separate M-variables can be used to access individual bits or segments of the image word. Use of these smaller M-variables allows PMAC's efficient firmware to do the masking and logic necessary to pick out portions of the I/O word, rather than slower user program code.

PMAC Location of Image Words

The internal images reside in PMAC's memory as follows:

Open Memory — Standard PMAC

For a standard PMAC with no DPRAM on board, the image word will be in an otherwise unused double register in PMAC's own memory. There are 16 open registers at PMAC addresses $0770 to $077F that are set to zero automatically on power-up. There are 16 more open registers at PMAC addresses $07F0 to $07FF, whose values are held when power is off. It is possible to use the registers of otherwise unused P and Q-variables for this purpose.

Access these registers with fixed-point M-variables, not floating-point P or Q-variables. A double fixed-point register in PMAC's internal memory is defined by the D format of M-variable (e.g. M61->D:$07F0). This is a 48-bit register (only the low 32 bits will be used). The low 24 bits of the I/O will be in the Y-memory, and the high eight bits of the I/O will be in the low eight bits of X-memory.

DPRAM — Standard PMAC

If the system has dual-ported RAM, use a 32-bit register in DPRAM. This way, the host computer always has immediate access to the I/O. In fact, PMAC can be used just as a pass-through between the host computer and the ACC-34x boards, letting the host computer do all the processing. A 32-bit fixed-point register in DPRAM is defined by the DP format of M-variable (e.g. M60-> DP:$DF00). This type of variable occupies the low 16 bits (bits 0 to 15) of PMAC Y-memory, and the low 16 bits of PMAC X-memory at the same address, with the less significant bits in Y-memory. It appears to the host computer as two 16-bit registers at consecutive even addresses, with the less significant bits at the lower address.

Turbo PMAC Location of Image Words

Open Memory Turbo PMAC

For a Turbo PMAC with no DPRAM on board, the image word will be in an unused double register in Turbo PMAC's own memory. There are 16 open registers at Turbo PMAC addresses $0010F0 to $0010FF that are set to zero automatically on power-up. Also, use the registers of otherwise unused P and Q-variables for this purpose.

Access these registers with fixed-point M-variables, not floating-point P or Q-variables. A double fixed-point register in PMAC's internal memory is defined by the D format of M-variable (e.g., M80->D:$0010F0). This is a 48-bit register (only the low 32 bits will be used). The low 24 bits of the I/O will be in the Y-memory, and the high 8 bits of the I/O will be in the low 8 bits of X-memory.

When using the ACC-34x I/O with fixed-point image variables, the only software overhead is the actual copying between image and I/O. Including program interpretation time, this amounts to about 100 microseconds per 32-bit word. Aside from this, working with the I/O through the image words is at least as fast as direct (parallel) PMAC I/O. There is a potential latency of a full PLC scan on the actual I/O, which must be respected. Many systems will have a few critical I/O points that cannot tolerate this latency; typically these use PMAC's JOPTO port or ACC-14 I/O for these time-critical points, then use ACC-34x for I/O that do not need to be so fast.

DPRAM Standard Turbo PMAC

If the system has dual-ported RAM (Option 2 is required), use a 32-bit register in DPRAM. This way, the host computer always has immediate access to the I/O. In fact, PMAC can be used just as a pass-through between the host computer and the ACC-34x boards, letting the host computer do all the processing. A 32-bit fixed-point register in DPRAM is defined by the DP format of M-variable (e.g. M80-> DP:$060000). This type of variable occupies the low 16 bits (bits 0 to 15) of PMAC Y-memory, and the low 16 bits of PMAC X-memory at the same address, with the less significant bits in Y-memory. It appears to the host computer as two 16-bit registers at consecutive even addresses, with the less significant bits at the lower address.

Preventing Conflicts in Output Image Words

Care must be taken if tasks of different priority levels are trying to write to the same output image word, or if both the host computer and PMAC are trying to write to the same DPRAM output image word. If the proper techniques are not used, occasional output changes will not be executed, and because of the intermittent nature of the problem, it will be very difficult to diagnose. If the application has two priority levels or two computers that write to the same ACC-34x output word, separate partial image words must be used. These words combined as the output word is sent.

Note

There is no conflict in having different tasks or different processors read from the same input word.

Remember that a computer cannot actually write to less than a word of memory at a time, even if it only wants to change one bit. In PMAC, the word length is 24 bits; for the DPRAM, it is 16 bits. If a computer wants to change less than a full word, it must read the full word, modify the bits it wants with mask words, then write back the full word.

There are two priority levels in PMAC that can write to these image words: the foreground level, which includes all of the motion programs and PLC 0; and the background level, which includes PLCs 1-31 and on-line commands. The problem can occur when a higher priority task interrupts a lower priority task that is in the middle of changing the image word with a read-modify-write operation. When the lower priority task resumes, it will undo the changes made by the higher priority task. Similarly, if the image word is in the DPRAM, and one side starts its read-modify-write cycle on the word but does not finish it before the other side starts its own cycle, the side that starts later can undo the changes made by the side that starts first.

Note

Two tasks at the same priority level cannot interrupt each other; one will always finish an operation before the other starts. Therefore, there is no need to worry about two motion programs writing to the same image word; or a motion program and PLC 0; because these tasks are at the same priority level. Similarly, there is no need to worry about two background PLC programs writing to the same image word, or a background PLC and on-line commands.)

To prevent this possible conflict, the different priority levels or different processors must use different image words, even if they each represent only a part of the same total output word. These partial words are then combined in the act of writing to the actual output word.

The simplest way to split an image word is to use the natural X-memory vs. Y-memory split in PMAC's memory. If using a double word in PMAC's internal memory, reserve the 24 bits in Y-memory for one priority level, and the eight bits in X-memory for the other. If using the DPRAM, reserve the 16 bits in Y-memory for one processor or priority level, and the 16 bits in X-memory for another. If using the memory way, no special techniques need to be used. On PMAC, simply write to the partial words with a X or Y format M-variable; PMAC will do the read-modify-write cycle automatically without touching the other part of the word. On the host computer, access the DPRAM register with the short (16-bit) integer format, not the long.

However, if the split cannot be arranged in this fashion, create separate overlapping image words and explicitly combine them.

Example: Take a system in which the low 12 bits will be written to by background PLCs and the high 20 bits will be written to by motion programs. Create two separate image words: one for each priority level, and the actual output word:

PMAC	TURBO PMAC	Comments
M101->D:$0770	M81->D:$0010F0	Image word for PLC programs (background)
M102->D:$0771	M82->D:$0010F1	Image word for motion programs (foreground)
M103->TWS:6	M83->TWS:6	ACC-34x output word; write-only

Also, single-bit M-variables are defined to parts of these same internal addresses: at Y:$0770 ($0010F0 for Turbo), bits 0 to 11 for the PLCs; then at Y:$0771($0010F1 for Turbo), bits 12 to 23, and

X:$0771($0010F1 for Turbo), bits 0 to 7 for the motion programs. At the end of a PLC scan, to create the actual output word on an ACC-34x from the image words, we would use the program statement:

```
M103 = (M101 & $00000FFF) | (M102 & $FFFFF000)
```

The bit-by-bit AND (&) operations make sure no falsely set bits in unused portions of the image words get into the output word. They are not strictly necessary if the unused bits can be guaranteed to be zero. The bit-by-bit OR (|) operation combines the word. The assignment of the resulting value to M103 causes it to be written to the ACC-34x.

To write to the same bit of an output image word with two different priority levels or processors, one of the tasks must do so indirectly by writing into a holding register. The other task must take this holding register and transfer the bit value into the image word. This task must decide what to do in case of any conflict (i.e., one task wants to clear the bit, and the other wants to set it).

The following example illustrates the method of working with ACC-34x I/O. It describes the procedure for memory allocation, for the inputs, and for the outputs (Image Word Variables) that will work with either Dual Ported RAM or PMAC memory locations

Example: This example shows the image variables both in DPRAM and several places in internal memory. In a real application, a single location range probably would be chosen.

Set-up and Definitions	Actual ACC-34 I/O Words
`M1000->TWS:1`	First side of first ACC-34x board; an input here Location is at port address 0; added 1 for read only
`M1002->TWS:6`	Second side of first ACC-34x board; an output here Location is at port address 4 added 2 for write only
`M1004->TWS:9`	First side of second ACC-34x board; an input here Location is at port address 8; added 1 for read only
`M1006->TWS:14`	Second side of second ACC-34x board; an output here Location is at port address 12; added 2 for write only

Image Words

PMAC	Turbo PMAC	Comments
M1001->DP:$D800	M1001->DP:$060800	32-bit fixed-point DPRAM register
M1003->D:$0770	M1003->D:$0010F0	48-bit fixed-point register, set to zero on power-up
M1005->D:$07F0	* Power-down hold registers are not available in Turbo PMAC M1005->D:$0061F0	PMAC: 48-bit fixed-point register, value held through power-down Turbo PMAC: use this register for P-variable, treated as 48-bit fixed point value
M1007->D:$13FF	M1007->D:$0063FF	Register for P1023, treated as 48-bit fixed-point value
* User Buffer Storage Space is same for the battery-backed Turbo PMAC.		

Individual Pieces of Image Words

PMAC Memory Locations

Port A Setup: Read Only		
PMAC	**Turbo PMAC**	**Comments**
M1000->TWS:1	M1000->TWS:1	Port A (AIO 0-31)
M1001->D:$770	M1003->D:$0010F0	Image word for PLCs
M800->Y:$770,0	M800->Y:$0010F0,0	Bit0 (LSB)
M801->Y:$770,1	M801->Y:$0010F0,1	Bit1
M802->Y:$770,2	M802->Y:$0010F0,2	Bit2
M822->Y:$770,22	M822->Y:$0010F0,22	Bit22
M823->Y:$770,23	M823->Y:$0010F0,23	Bit23
M824->X:$770,0	M824->X:$0010F0,0	Bit24
M825->X:$770,1	M825->X:$0010F0,1	Bit25
M830->X:$770,6	M830->X:$0010F0,6	Bit30
M831->X:$770,7	M831->X:$0010F0,7	Bit31 (MSB)
Port B Setup: Write Only		
PMAC	**Turbo PMAC**	**Comments**
M1002->TWS:6	M1002->TWS:6	Port B (BIO 0-31)
M1003->D:$771	M1003->D:$0010F1	Image word for PLCs
M900->Y:$771,0	M900->Y:$0010F1,0	Bit0 (LSB)
M901->Y:$771,1	M901->Y:$0010F1,1	Bit1
902->Y:$771,2	M902->Y:$0010F1,2	Bit2
922->Y:$771,22	M922->Y:$0010F1,22	Bit22
923->Y:$771,23	M923->Y:$0010F1,23	Bit23
924->X:$771,0	M924->X:$0010F1,0	Bit24
925->X:$771,1	M925->X:$0010F1,1	Bit25
M930->X:$771,6	M930->X:$0010F1,6	Bit30
M931->X:$771,7	M931->X:$0010F1,7	Bit31 (MSB)

PMAC Dual Ported RAM Locations

Port A Setup: Read Only		
PMAC	**Turbo PMAC**	**Comments**
M1000->TWS:1	M1000->TWS:1	Port A (AIO 0-31)
M1001->DP:$D800	M1001->DP:$060800	Image word for PLCs
M800->Y:$D800,0	M800->Y:$060800,0	Bit0 (LSB)
M801->Y:$D800,1	M801->Y:$060800,1	Bit1
M802->Y:$D800,2	M802->Y:$060800,2	Bit2
M814->Y:$D800,14	M814->Y:$060800,14	Bit14
M815->Y:$D800,15	M815->Y:$060800,15	Bit15
M816->X:$D800,0	M816->X:$060800,0	Bit16
M817->X:$D800,1	M817->X:$060800,1	Bit17
M830->X:$D800,14	M830->X:$060800,14	Bit30
M831->X:$D800,15	M831->X:$060800,15	Bit31 (MSB)
Port B Setup: Write Only		
PMAC	**Turbo PMAC**	**Comments**
M1002->TWS:6	M1002->TWS:6	Port B (BIO 0-31)
M1003->DP:$D801	M1003->DP:$060801	Image word for PLCs
M900->Y:$D801,0	M900->Y:$060801,0	Bit0 (LSB)
M901->Y:$D801,1	M901->Y:$060801,1	Bit1
M902->Y:$D801,2	M902->Y:$060801,2	Bit2

M914->Y:$D801,14	M914->Y:$060801,14	Bit14
M915->Y:$D801,15	M915->Y:$060801,15	Bit15
M916->X:$D801,0	M916->X:$060801,0	Bit16
M917->X:$D801,1	M917->X:$060801,1	Bit17
M930->X:$D801,14	M930->X:$060801,14	Bit30
M931->X:$D801,15	M931->X:$060801,15	Bit31 (MSB)

```
Programs:              ; Reset PLC program that only runs once on
                       ; power-up or reset
OPEN PLC 1 CLEAR
M1003=0                ; Clear output image word to make sure all
                       ; outputs off
M1007=0                ; Ditto
...
DISABLE PLC 1          ; To make sure this only runs once on
                       ; power-up/reset
CLOSE
                       ; PLC program to copy the inputs into image
                       ; words at beginning of each scan

OPEN PLC 2 CLEAR
M1001=M1000            ; Copy first input word into its image
                       ; register
M1005=M1004^$FFFFFFFF  ; Copy second input word into its image
                       ; register, inverting
...
CLOSE
                       ; PLC program that works with individual
                       ; bits of image  words

OPEN PLC 3 CLEAR
IF (M100=1 AND M101=0 AND P43>50)
    M301=1
ELSE
    M301=0
ENDIF
...
CLOSE
                       ; PLC program that copies image words to
                       ; outputs at end of scan

OPEN PLC 31 CLEAR
M1002=M1003            ; Copy first output image word to ACC-34x
M1004=M1005^$FFFFFFFF  ; Copy second output image word to ACC-34x,
                       ; inverting
```

USING ACC-34 WITH MACRO STATION

Two ACC-34s could be used at each MACRO Station in the ring. The MACRO Station has eight variables used for this purpose: MI90, MI91, MI92, MI93, MI94, MI95, MI96, and MI97. Each ACC-34 Port has two variables associated with it. The first variable is the multiplexer port address, and the second is the actual input or output word. To read or write to these variables, special MACRO Station read and write commands (buffered and on-line) can be used to read and write to and from the Ultralite or Turbo Ultralite.

Port Address	Actual Word	Type
MI90	MI91	Input – Read Only
MI92	MI93	Output – Write Only
MI94	MI95	Input – Read Only
MI96	MI97	Output – Write Only

These variables are implemented on MACRO Station firmware versions V1.111 and newer.

Example: To connect two ACC-34AAAs to the multiplexer port on the MACRO station for 128-bits of I/O. The SW1 settings for the boards in this example are shown below.

Board #	Byte (Port A & Port B)	5	4	3	2	1
1	0 & 4	ON	ON	ON	ON	ON
2	8 & 12	ON	ON	ON	ON	OFF

Based on the switch settings, the port address settings would be:

```
MSn,MI90=1        ; Port A Board 1
MSn,MI92=6        ; Port B Board 1
MSn,MI94=9        ; Port A Board 2
MSn,MI96=14       ; Port B Board 2
```

Read and write from and to the MACRO station using the following definitions and commands:

PMAC Ultralite	Turbo PMAC Ultralite	Comments
M1000->D:$07F0	M1000->D:$0061F0	Input image word for Ultralite board 1
M1001->D:$07F1	M1001->D:$0061F1	Output image word for Ultralite board 1
M1002->D:$07F2	M1002->D:$0061F2	Input image word for Ultralite board 2
M1003->D:$07F3	M1003->D:$0061F3	Output image word for Ultralite board 2

Use the following commands as either on-line commands for testing, or in PLC or motion program as buffer commands:

```
MSRn,MI91,M1000       ; Reads the value of MSn, MI91 to M1000
MSWn,MI93,M1001       ; Writes the value of M1001 to MSn, MI93
MSRn,MI95,M1002       ; Reads the value of MSn, MI95 to M1002
MSRn,MI97,M1003       ; Reads the value of M1003 to MSn, MI97
```

MS{anynode},MI90: Multiplexer Port #1 Read Address

Range: $00 - $FF
Units: MACRO Station Multiplexer Port Addresses
Default: $00

MI90 specifies the address on the MACRO Station's JTHW multiplexer port of a 32-bit input port on an ACC-34 family I/O board. If MI90 is set greater than 0, the 32 input values will be copied periodically into Station variable MI91. MI90 must match the multiplexer port address of the ACC-34 board from which the inputs are to be read, as set by the SW1 DIP switch bank on the board. For the regular input port, the value of MI90 should be set one greater than the base address of the board set by the DIP switch bank. For the ACC-34C's optional second 32-bit input port, the value of MI94 should be set three greater than the base address.

MS{anynode},MI91: Multiplexer Port #1 Read Value

Range: $00000000 - $FFFFFFFF
Units: Individual bits

MI91 contains the 32-bit value read from the input port of the ACC-34 whose multiplexer port address is specified by MI90. Each bit represents one input from the port. Bit *n* of MI91 represents Input *n* on the port.

MS{anynode},MI92: Multiplexer Port #1 Write Address

Range: $00 - $FF
Units: MACRO Station Multiplexer Port Addresses
Default: $00

MI92 specifies the address on the MACRO Station's JTHW multiplexer port of a 32-bit output port on an ACC-34 family I/O board. If MI92 is set greater than 0, the 32 output values will be copied periodically from Station variable MI93. MI92 must match the multiplexer port address of the ACC-34 board from which the inputs are to be read, as set by the SW1 DIP switch bank on the board. The value of MI92 should be set 6 greater than the base address of the board set by the DIP switch bank.

MS{anynode},MI93: Multiplexer Port #1 Write Value

Range: $00000000 - $FFFFFFFF
Units: Individual bits

MI93 contains the 32-bit value written to the output port of the ACC-34 whose multiplexer port address is specified by MI92. Each bit represents one output on the port. Bit *n* of MI93 represents Output *n* on the port.

MS{anynode},MI94: Multiplexer Port #2 Read Address

Range: $00 - $FF
Units: MACRO Station Multiplexer Port Addresses
Default: $00

MI94 specifies the address on the MACRO Station's JTHW multiplexer port of a 32-bit input port on an ACC-34 family I/O board. If MI94 is set greater than 0, the 32 input values will be copied periodically into Station variable MI95. MI94 must match the multiplexer port address of the ACC-34 board from which the inputs are to be read, as set by the SW1 DIP switch bank on the board. For the regular input port, the value of MI94 should be set one greater than the base address of the board set by the DIP switch bank. For the ACC-34C's optional second 32-bit input port, the value of MI94 should be set three greater than the base address.

MS{anynode},MI95: Multiplexer Port #2 Read Value

Range: $00000000 - $FFFFFFFF
Units: Individual bits

MI95 contains the 32-bit value read from the input port of the ACC-34 whose multiplexer port address is specified by MI94. Each bit represents one input from the port. Bit *n* of MI95 represents Input *n* on the port.

MS{anynode},MI96: Multiplexer Port #2 Write Address

Range: $00 - $FF
Units: MACRO Station Multiplexer Port Addresses
Default: $00

MI96 specifies the address on the MACRO Station's JTHW multiplexer port of a 32-bit output port on an ACC-34 family I/O board. If MI96 is set greater than 0, the 32 output values will be copied periodically from Station variable MI97. MI96 must match the multiplexer port address of the ACC-34 board from which the inputs are to be read, as set by the SW1 DIP switch bank on the board. The value of MI96 should be set six greater than the base address of the board set by the DIP switch bank.

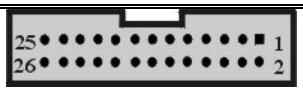
MS{anynode},MI97: Multiplexer Port #1 Write Value

Range: $00000000 - $FFFFFFFF
Units: Individual bits

MI97 contains the 32-bit value written to the output port of the ACC-34 whose multiplexer port address is specified by MI96. Each bit represents one output on the port. Bit *n* of MI97 represents Output *n* on the port.

CONNECTOR PINOUTS

J1A and J1B (26-Pin Header)

Top View 

Pin #	Symbol	Function	Description	Notes
1	GND	Common	PMAC Common	
2	GND	Common	PMAC Common	
3	DAT0	Output	Data Bit 0	
4	SEL0	Input	Address Line 0	
5	DAT1	Output	Data Bit 1	Not connected
6	SEL 1	Input	Address Line 1	
7	DAT2	Output	Data Bit 2	Not connected
8	SEL2	Input	Address Line 2	
9	DAT3	Output	Data Bit 3	Not connected
10	SEL3	Input	Address Line 3	
11	DAT4	Output	Data Bit 4	Not connected
12	SEL 4	Input	Address Line 4	
13	DAT5	Output	Data Bit 5	Not connected
14	SEL5	Input	Address Line 5	
15	DAT6	Output	Data Bit 5	Not connected
16	SEL6	Input	Address Line 6	
17	DAT7	Output	Data Bit 6	Not connected
18	SEL7	Input	Data Bit 7	
19	N.C.			Not connected
20	GND	Common	PMAC Common	
21	N.C.			Not connected
22	GND	Common	PMAC Common	
23	N.C.			Not connected
24	GND	Common	PMAC Common	
25	+5V	Input	+5V DC Supply	
26	N.C.			Not connected

This header should be connected to PMAC's JTHW connector via the supplied 26-pin flat cable.

TB1 (10-pin Terminal Block)

Top View 

Pin #	Symbol	Function	Description	Notes
1	AI0	Input	Port A Bit 0	
2	AI1	Input	Port A Bit 1	
3	AI2	Input	Port A Bit 2	
4	AI3	Input	Port A Bit 3	
5	AI4	Input	Port A Bit 4	
6	AI5	Input	Port A Bit 5	
7	AI6	Input	Port A Bit 6	
8	AI7	Input	Port A Bit 7	
9	AGND	I/O Common	Return Signal	
10	AV+1	I/O Supply	15 to 24V	*

* Used only for the ten 3.3kς pull-up/down resistors and LEDs (15 to 24V, 70mA).

TB2 (10-pin Terminal Block)

Top View 

Pin #	Symbol	Function	Description	Notes
1	AI8	Input	Port A Bit 8	
2	AI9	Input	Port A Bit 9	
3	AI10	Input	Port A Bit 10	
4	AI11	Input	Port A Bit 11	
5	AI12	Input	Port A Bit 12	
6	AI13	Input	Port A Bit 13	
7	AI14	Input	Port A Bit 14	
8	AI15	Input	Port A Bit 15	
9	AGND	I/O Common	Return Signal	
10	AV+2	I/O Supply	15 to 24V	*

* Used only for the ten 3.3kς pull-up/down resistors and LEDs (15 to 24V, 70mA).

TB3 (10-Pin Terminal Block)

Top View 

Pin #	Symbol	Function	Description	Notes
1	AI16	Input	Port A Bit 16	
2	AI17	Input	Port A Bit 17	
3	AI18	Input	Port A Bit 18	
4	AI19	Input	Port A Bit 19	
5	AI20	Input	Port A Bit 20	
6	AI21	Input	Port A Bit 21	
7	AI22	Input	Port A Bit 22	
8	AI23	Input	Port A Bit 23	
9	AGND	I/O Common	Return Signal	
10	AV+3	I/O Supply	15 to 24V	*

* Used only for the ten 3.3kς pull-up/down resistors and LEDs (15 to 24V, 70mA).

TB4 (10-Pin Terminal Block)

Top View 

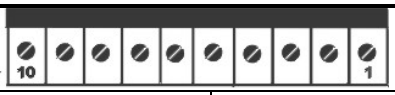
Pin #	Symbol	Function	Description	Notes
1	AI24	Input	Port A Bit 24	
2	AI25	Input	Port A Bit 25	
3	AI26	Input	Port A Bit 26	
4	AI27	Input	Port A Bit 27	
5	AI28	Input	Port A Bit 28	
6	AI29	Input	Port A Bit 29	
7	AI30	Input	Port A Bit 30	
8	AI31	Input	Port A Bit 31	
9	AGND	I/O Common	Return Signal	
10	AV+4	I/O Supply	15 to 24V	

* Used only for the ten 3.3kς pull-up/down resistors and LEDs (15 to 24V, 70mA)

TB5 (10-Pin Terminal Block)			Top View 	
Pin #	**Symbol**	**Function**	**Description**	**Notes**
1	BO0	Output	Port B Bit 0	
2	BO1	Output	Port B Bit 1	
3	BO2	Output	Port B Bit 2	
4	BO3	Output	Port B Bit 3	
5	BO4	Output	Port B Bit 4	
6	BO5	Output	Port B Bit 5	
7	BO6	Output	Port B Bit 6	
8	BO7	Output	Port B Bit 7	
9	AGND	I/O Common	Return Signal	
10	AV+5	I/O Supply	12 to 24V	*
* In the sourcing configuration, this supply will be used to drive the loads on all eight lines (up to 800mA maximum at 24V). In the sinking configuration, the current requirement from this power supply is minimal (approximately 70mA).				

TB6 (10-Pin Terminal Block)			Top View 	
Pin #	**Symbol**	**Function**	**Description**	**Notes**
1	BO8	Output	Port B Bit 8	
2	BO9	Output	Port B Bit 9	
3	BO10	Output	Port B Bit 10	
4	BO11	Output	Port B Bit 11	
5	BO12	Output	Port B Bit 12	
6	BO13	Output	Port B Bit 13	
7	BO14	Output	Port B Bit 14	
8	BO15	Output	Port B Bit 15	
9	AGND	I/O Common	Return Signal	
10	AV+6	I/O Supply	12 to 24 Volts	*
* In the sourcing configuration, this supply will be used to drive the loads on all eight lines (up to 800mA maximum at 24V). In the sinking configuration, the current requirement from this power supply is minimal (approximately 70mA).				

TB7 (10-Pin Terminal Block)

Top View 

Pin #	Symbol	Function	Description	Notes
1	BO16	Output	Port B Bit 16	
2	BO17	Output	Port B Bit 17	
3	BO18	Output	Port B Bit 18	
4	BO19	Output	Port B Bit 19	
5	BO20	Output	Port B Bit 20	
6	BO21	Output	Port B Bit 21	
7	BO22	Output	Port B Bit 22	
8	BO23	Output	Port B Bit 23	
9	AGND	I/O Common	Return Signal	
10	AV+7	I/O Supply	12 to 24 Volts	*

* In the sourcing configuration, this supply will be used to drive the loads on all eight lines (up to 800 mA maximum at 24V). In the sinking configuration, the current requirement from this power supply is minimal (approx. 70 mA).

TB8 (10-Pin Terminal Block)

Top View 

Pin #	Symbol	Function	Description	Notes
1	BO24	Output	Port B Bit 24	
2	BO25	Output	Port B Bit 25	
3	BO26	Output	Port B Bit 26	
4	BO27	Output	Port B Bit 27	
5	BO28	Output	Port B Bit 28	
6	BO29	Output	Port B Bit 29	
7	BO30	Output	Port B Bit 30	
8	BIO31	Output	Port B Bit 31	
9	AGND	I/O Common	Return Signal	
10	AV+8	I/O Supply	5 to 24 Volts	*

* In the sourcing configuration, this supply will be used to drive the loads on all eight lines (up to 800 mA maximum at 24V). In the sinking configuration, the current requirement from this power supply is minimal (approx. 70 mA).

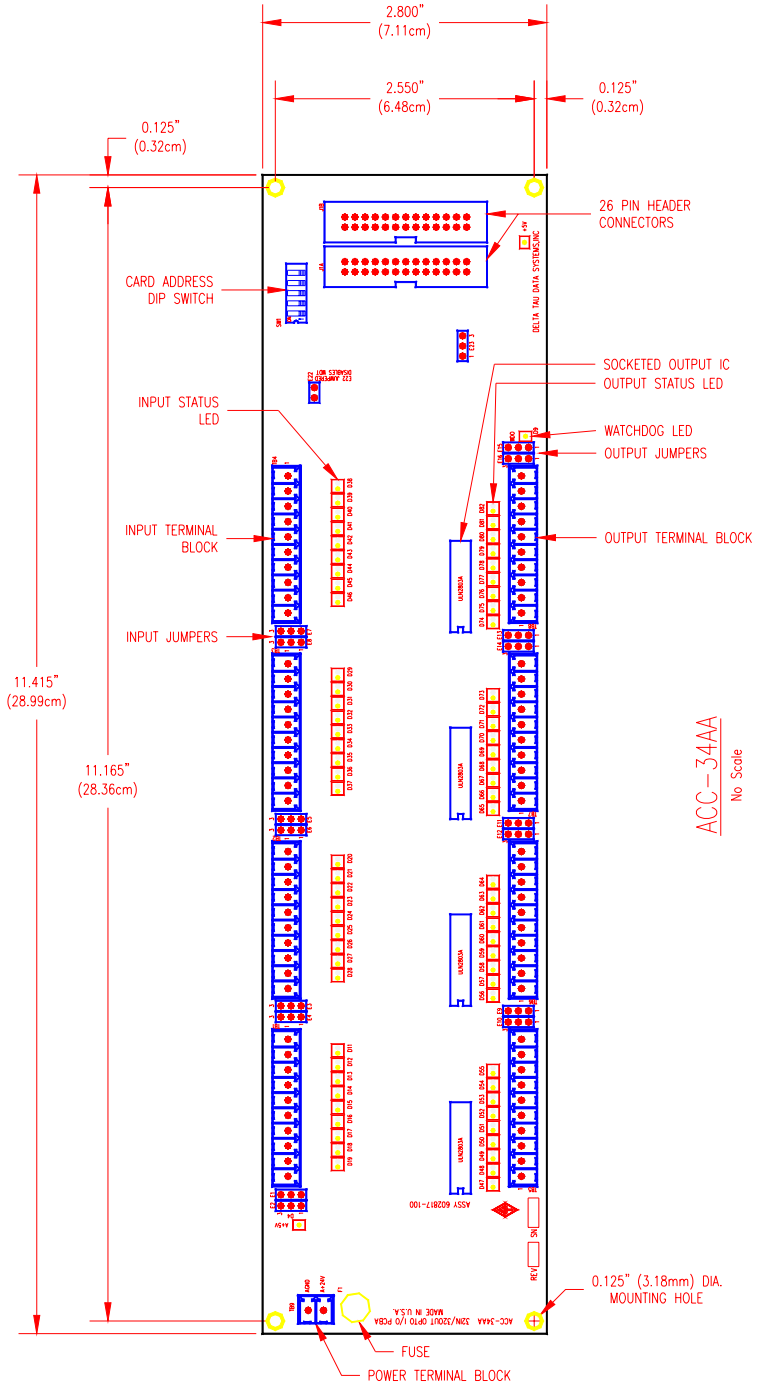
TB9 (2-Pin Terminal Block)

Top View 

Pin #	Symbol	Function	Description	Notes
1	AGND	Common	External Supply Ground	
2	A+24V	Power Supply	External Supply for Opto-Isolators	12V to 24 V Unregulated

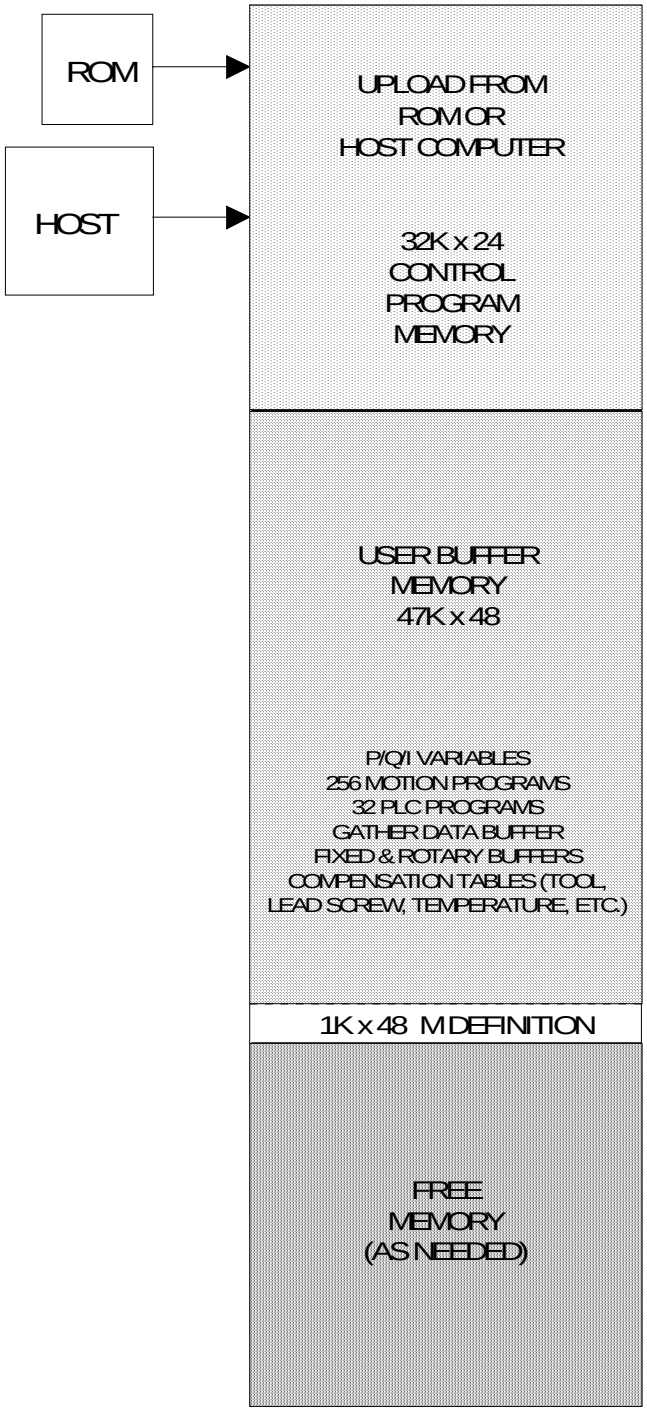
TB9 is a 2-pin terminal block used to bring in the power supply (12-24V) for the I/O side of the opto-isolation circuitry. For proper operation of the board, this power supply must be brought in through TB9.

MECHANICAL DIMENSIONS



MEMORY MAP

PMAC Memory



ROM = 128K x 8

EAROM = 2K x 8
Stores I Parameters only

RAM = 128K X 24
Battery Backed
Program, X & Y Memory

RAM:
NORMAL: 65 nsecs
1 WAIT STATE/20 MHz

MEDIUM: 45 nsecs
0 WAIT STATE/20 MHz

FAST: 25 nsecs
0 WAIT STATE/30 MHz

End of RAM is definable by user